

ETC5513_Assignment1

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Introduction

Marketing plays a crucial role in driving customer engagement and sales growth. However, the effectiveness of marketing spend remains uncertain. Therefore, using the data set selected, this research aims to look into the relationship between marketing investment and customer purchase behavior in order to gather insights for future budget allocation and campaign strategy.

Research Question

This report investigates whether higher marketing spend per customer is associated with higher purchase amounts. It also examines whether this relationship differs across customer segments(Enterprise, SMB, and Consumer), in order to compare how the strength of the relationship varies between segments.

Dataset Introduction

Step 1: Modify the date format

Step 2: Reduce observations with missing purchase amount values

This data set contains 132 observations of individual customers. The data contains customers' purchase amounts, number of visits, and segment classifications, as well as the marketing spend allocated to individual customers. It captures both customer engagement and purchasing behavior, which could be used to analyze the relationship between marketing investment and customer purchase behavior in this research.

Dataset Description

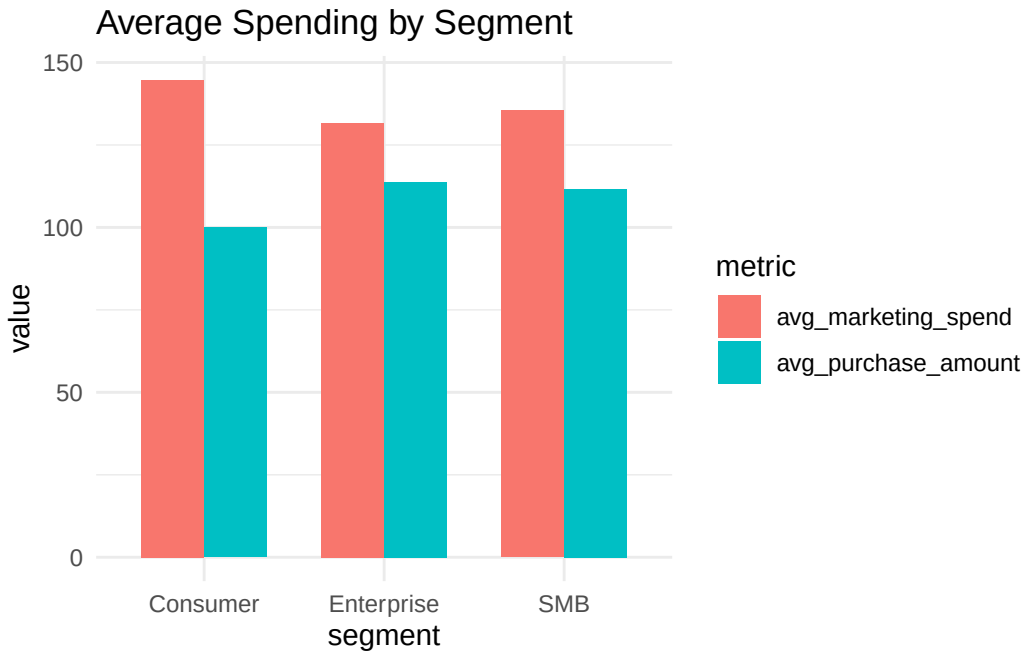
This dataset contains 132 observations and 7 variables. It includes `customer_id`, `signup_date`, `marketing_spend`, `visits`, `purchase_amount`, `segment` and `notes`. Among these, `signup_date` is a date variable, `customer_id`, `segment`, and `notes` are character variables, and `marketing_spend`, `visits`, and `purchase_amount` are numeric variables.

- Data Summary

The following table provides an overview of the minimum and maximum purchase and marketing spending across different segments, while the plot shows the average spending across segments.

Table 1: Segment Summary

segment	n	min_purchase	max_purchase	min_mkting	max_mkting
Consumer	93	0	232.56	3.18	294.89
Enterprise	9	0	272.19	5.10	288.44
SMB	30	0	282.95	13.30	277.31



Results

From Figure 1, the scatter plot shows an upward trend, suggesting a positive relationship between marketing spend and purchase amount. Further evidence is provided by the linear regression model, where the p-value for marketing spend is less than 0.001, indicating that the relationship is statistically significant. Therefore, it is likely that higher marketing spend is associated with higher purchase amounts.

Figure 2 illustrates the distribution and fitted regression line across different segments. A consistent upward trend is observed in all groups, confirming a statistically significant positive correlation between marketing spend and purchase amount. While the direction of the relationship remains uniform, the magnitude of the slopes vary by segments: Consumer(0.475), Enterprise(0.743), and SMB(0.740). Specifically, the Enterprise segment exhibits the steepest gradient, suggesting that incremental marketing investments yield the highest marginal returns in purchase volume within the group.

These regression results also align with the findings from the data summary. Although the Consumer segment has the highest average marketing spend, it yields lowest purchase amount among all groups. This discrepancy can be explained by its lowest regression slope(0.475), which indicates that each unit of marketing investment in this segment generates the smallest incremental increase in revenue.

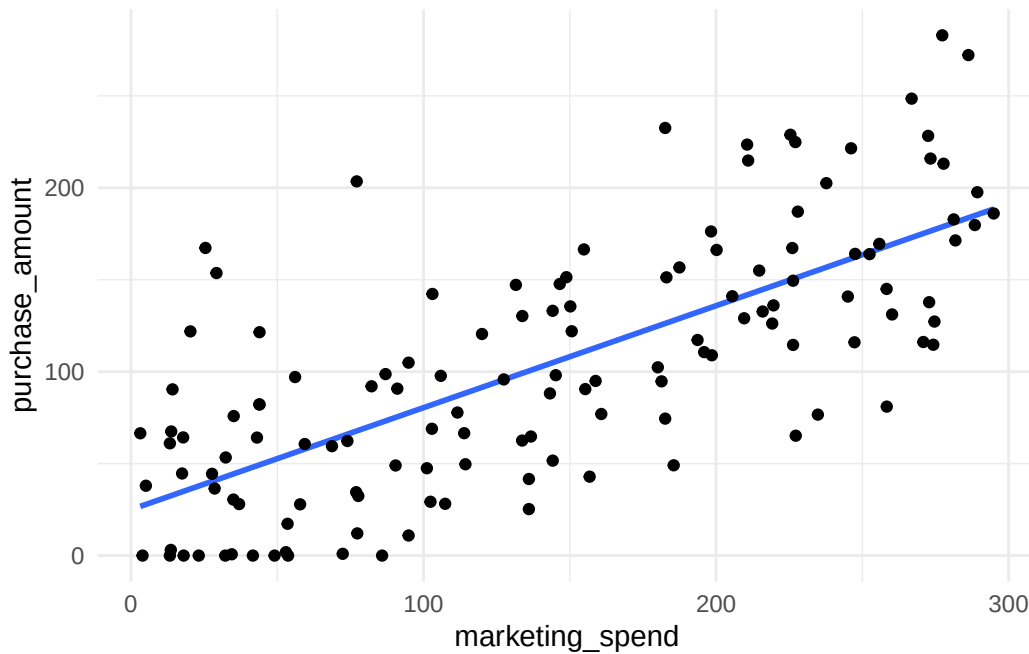


Figure 1: Marketing Spend vs. Purchase Amount

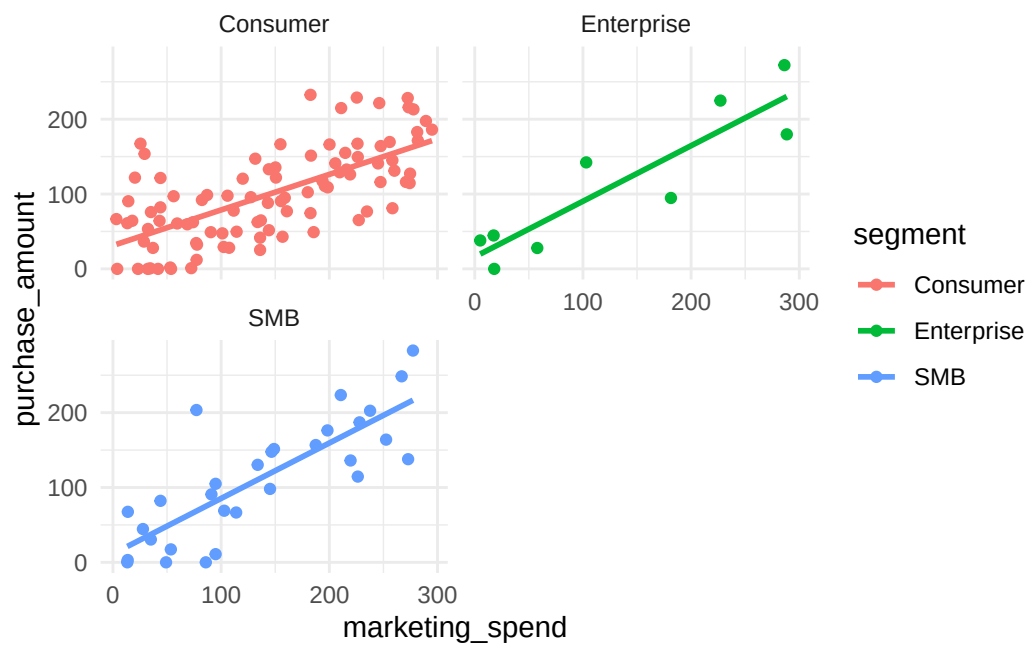


Figure 2: Marketing Spend vs. Purchase Amount Across Segments